

CAN Bus Assignment Part 1

Technical Report

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Introduction

This assignment focuses on the CAN bus protocol. For part 1, two Arduino boards are communicating with each other via the CAN bus, using the mcp2515 modules and their library functions. We must send a command from a **transmitter** Arduino to a **receiver** Arduino and control a LED.

Hardware

- 2 Arduino Uno R3 based Sparkfun Redboards
- 2 mcp2515 CAN modules
- 1 LED

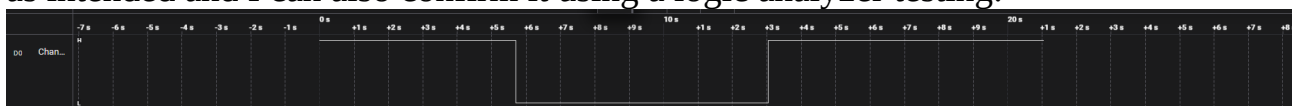
Software

The software uses the Arduino abstraction layer for handling embedded functions. Using the mcp2515 library to interface with the can modules.

There are two Arduinos at play, the “transmitter” and the “receiver”. The transmitter, at address 0x43, sends user commands via the CAN bus to the receiver. The receiver interprets the command and executes an action on its end to toggle a LED. If the command is “on” the receiver turns the light on, otherwise it turns it off and if it encounters a wrong command, it ignores it.

Testing and measurements

By visually confirming I could, with confidence tell, that the program is functioning as intended and I can also confirm it using a logic analyzer testing.



You can see the period where I typed “off” turning off the LED and turning it on again using “on”.

Note: A bug in the debugging serial monitor logs I placed dictate a failure when sending the payload. I have not yet found out why exactly it happens, but the program still functions normally so I will not put effort into fixing, for now.

Conclusion

The program is functioning as intended, exception is the logging bug, but it does not affect functionality.